

Express Riddler

4 March 2022

Riddle:

You have three fair coins, three fair dice and a full deck of cards in your possession. First, you flip all three coins and note the number of heads. Next, you toss all three dice and note the number of ones or sixes. Finally, you draw three random cards from the deck of 52 and note the number of hearts.

What is the probability that all three numbers are the same?

Solution:

The probabilities of flipping zero, one, two, and three heads are

$$\begin{aligned}P(0 \text{ head}) &= \left(\frac{1}{2}\right)^3 = \frac{1}{8} \\P(1 \text{ head}) &= 3 \left(\frac{1}{2}\right)^3 = \frac{3}{8} \\P(2 \text{ head}) &= 3 \left(\frac{1}{2}\right)^3 = \frac{3}{8} \\P(3 \text{ head}) &= \left(\frac{1}{2}\right)^3 = \frac{1}{8}\end{aligned}$$

The probabilities of rolling zero, one, two, and three 1s/6s are

$$\begin{aligned}P(0 \text{ 1/6}) &= \left(\frac{2}{3}\right)^3 = \frac{8}{27} \\P(1 \text{ 1/6}) &= 3 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right) = \frac{4}{9} \\P(2 \text{ 1/6}) &= 3 \left(\frac{2}{3}\right) \left(\frac{1}{3}\right)^2 = \frac{2}{9} \\P(3 \text{ 1/6}) &= \left(\frac{1}{3}\right)^3 = \frac{1}{27}\end{aligned}$$

I make the assumption that the cards are drawn without replacement, so that the probability of each draw changes. Then the probabilities of drawing zero, one, two, and three hearts are

$$\begin{aligned}P(0 \text{ heart}) &= \left(\frac{39}{52}\right) \left(\frac{38}{51}\right) \left(\frac{37}{50}\right) = \frac{703}{1700} \\P(1 \text{ heart}) &= \left(\frac{13}{52}\right) \left(\frac{39}{51}\right) \left(\frac{38}{50}\right) + \left(\frac{39}{52}\right) \left(\frac{13}{51}\right) \left(\frac{38}{50}\right) + \left(\frac{39}{52}\right) \left(\frac{38}{51}\right) \left(\frac{13}{50}\right) = \frac{741}{1700} \\P(2 \text{ heart}) &= \left(\frac{13}{52}\right) \left(\frac{12}{51}\right) \left(\frac{39}{50}\right) + \left(\frac{13}{52}\right) \left(\frac{39}{51}\right) \left(\frac{12}{50}\right) + \left(\frac{39}{52}\right) \left(\frac{13}{51}\right) \left(\frac{12}{50}\right) = \frac{117}{850} \\P(3 \text{ heart}) &= \left(\frac{13}{52}\right) \left(\frac{12}{51}\right) \left(\frac{11}{50}\right) = \frac{11}{850}\end{aligned}$$

The probabilities of getting all zeroes, ones, twos, and threes are the products of the respective probabilities:

$$\begin{aligned}P_0 &= P(0 \text{ head})P(0 \text{ 1/6})P(0 \text{ heart}) = \frac{2,812}{183,600} \\P_1 &= P(1 \text{ head})P(1 \text{ 1/6})P(1 \text{ heart}) = \frac{13,338}{183,600} \\P_2 &= P(2 \text{ head})P(2 \text{ 1/6})P(2 \text{ heart}) = \frac{2,106}{183,600} \\P_3 &= P(3 \text{ head})P(3 \text{ 1/6})P(3 \text{ heart}) = \frac{11}{183,600}\end{aligned}$$

The total probability is then the sum:

$$P = P_0 + P_1 + P_2 + P_3 = \frac{18,267}{183,600}$$

So the solution is $\frac{18,267}{183,600} \approx 0.0995$.